Test Cases & Test Report

HDI Prediction System | Project Testing Phase

Testing Scope

Testing covers the prediction pipeline in app.py: input validation, the classify_hdi() tier boundaries, both the form-based and JSON API code paths, and graceful error handling.

Test Cases

ID	Test Description	Input (life, expected, mean, income)	Expected Result
TC-01	Valid input, Very High tier	80.0, 16.0, 12.5, 45000	HDI score clipped to [0,1]; tier = 'Very High'
TC-02	Valid input, Medium tier	63.0, 9.5, 6.0, 4500	Tier = 'Medium'
TC-03	Valid input, Low tier	50.0, 6.0, 3.0, 900	Tier = 'Low'
TC-04	Boundary value at High threshold	score approx 0.700	Tier = 'High' (>= 0.700 inclusive)
TC-05	Boundary value at Very High threshold	score approx 0.800	Tier = 'Very High' (>= 0.800 inclusive)
TC-06	Missing field (income omitted)	70.0, 12.0, 8.0, <missing>	Returns friendly error: 'Please provide valid numeric va
TC-07	Non-numeric input	'abc', 12.0, 8.0, 5000	Caught by ValueError/TypeError handling; error messa
TC-08	JSON API request	POST /predict with JSON body	Returns JSON: {score, tier_label, tier_class, inputs}
TC-09	Form-based browser request	POST /predict via HTML form	Renders index.html with prediction_text populated
TC-10	Out-of-range raw prediction (model extrapolation)	Extreme input values	Score is clamped between 0.0 and 1.0 before tier class

Test Environment

- Framework: Flask development server (debug=True) for local testing.
- Model: HDI.pkl loaded once at application startup.
- Dependencies: Flask, numpy, pandas, scikit-learn, gunicorn (as per requirements.txt).

Test Summary Report

ID	Status	Notes
TC-01	Pass	Score and tier returned as expected.
TC-02	Pass	Score and tier returned as expected.
TC-03	Pass	Score and tier returned as expected.
TC-04	Pass	Boundary confirmed inclusive at 0.700.
TC-05	Pass	Boundary confirmed inclusive at 0.800.
TC-06	Pass	Error message shown instead of server crash.

ID	Status	Notes
TC-07	Pass	Error message shown instead of server crash.
TC-08	Pass	Valid JSON structure returned with 200 status.
TC-09	Pass	Prediction text rendered correctly on index.html.
TC-10	Pass	Prediction clipped to valid [0, 1] range.

Conclusion

All planned test cases passed. The application correctly validates input, classifies predictions into the four official UNDP development tiers, and handles both browser-based and API-based requests without crashing on invalid input.